**Post-Disaster Damage Assessment and Rescue Guidance**

CSE499B – Section 13, Group 5, Update 4


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# Weekly Update:

Our system reads before-and-after satellite images of a disaster, marks the damage building by building, counts it, and writes a rescue report that cites official guidelines, offline, on hardware small enough for the field. This week the report stage passed its proof: on Kaggle, with the internet switched off, DisasTeller's report pipeline ran on our own fine-tuned Qwen2.5-VL-7B instead of Google's Gemini and wrote all six rescue reports, with no connection leaving the machine. It needed a change of design: the 7B could not choose its own tools, so the code now calls them in a fixed order and the model only writes. The proof shows the reports *can* be written offline, not yet that they are good; that comparison is next. We also began turning the work into something people can use: the first version of our website and demo (Noman), and a query console that answers from the building table, never from the VLM (Aryan). Abrar rewrote the project's front page around our four objectives and readied a second test set; Tamanna tested two new ideas, neither of which helped. Scores marked clean are on the 605 validation pairs left after removing 40 whose before-image also appears in training.

# Results Summary

| Goal | Target | Best result so far | Status |
|---|---|---|---|
| 1. Damage map | Mean IoU above 0.52 and destroyed-building IoU above 0.30, on two training seeds | Find-then-grade, clean pairs: destroyed-building IoU 0.316, 0.305 and 0.298 with three graders (mean 0.307); mean IoU 0.547 | Met on the mean of three graders, narrowly; second test set ready |
| 2. Counting | Above the base VLM's 34.2% on the benchmark's counting questions | 32.96% by a fixed query over a table of the buildings our U-Net found (32.47% without duplicated images) | 1.7 points short |
| 3. Offline report | The report-writing agents on our own models, with damage map, counts and rescue heatmap | Proof run passed with the internet off (11 Nov): our 7B wrote all six reports; 4 connections, all inside the machine; no Gemini | Offline proof met; report quality next |
| 4. Field model | A small segmenter and a distilled 3B VLM | U-Net in 8-bit integers: 33 MB, 1.43× faster on a CPU, destroyed IoU down 0.0003; 1,146 training answers for the 3B | Segmenter done; 3B training next |

IoU (intersection over union): the overlap between the pixels predicted as a label and the pixels truly of it, from 0 to 1; mean IoU averages it over background, intact, damaged and destroyed.

# Abrar Mohammed Tanzim Alam – Project Overview and System Status

1. **Project front page rewritten (11 Nov):** the README now explains, for a new reader, what the system gives a rescue coordinator: a damage map, building by building; counts from a table of the buildings, never guessed by a language model; a rescue-priority heatmap; and reports that cite EMS-98, INSARAG and Sphere, written by our own VLM offline. It draws the whole pipeline in one diagram, from SAM 3's outlines to the six reports and the field model, states where each objective stands on 11 Nov with every result linked to its notebook, and indexes all CSE499A and CSE499B updates and what each folder holds. Files: README.md
2. **A second test set:** 2,007 unseen image pairs are ready as a second test set. Find-then-grade passed the damage-map target only narrowly, on validation pairs that also decided which map we kept, so the next check is on pairs no choice was made on.
3. **Find-then-grade (R16c), our current damage map:** SAM 3 outlines every building on the before-image; a Siamese ResNet-18 compares each building's before and after crops and grades it intact, damaged, destroyed or not a building. On clean pairs its destroyed-building IoU is 0.316, 0.305 and 0.298 with three graders (mean 0.307) and its mean IoU 0.547. With perfect grades on SAM 3's buildings it would reach 0.556, so most of the remaining gap is the grader. Files: Champion_Task/R16c-results/step5b/README.md
4. **R11, the alternative under test:** the U-Net that reads both images has our best mean IoU (0.588) but a clean destroyed-building IoU of 0.269: Turkey's 2023 earthquake holds 37% of all destroyed pixels and R11 scores 0.000 there (0.410 without Turkey). Files: Champion_Task/R11-results/d4_R11/D4_R11.csv
5. **The VLM that writes the reports:** our Qwen2.5-VL-7B, trained 28 more steps at full image size (1036 px), was kept by the rule fixed before the run: relational reasoning +3.5 points ($p < 0.001$), disaster type +1.1, no other task down by more than 1 point; mean 38.4% → 39.2%. It is the model that wrote this week's offline reports; its counting (30.3%) is still below 34.2%, so counts come from the table. Files: Champion_Task/L0-step2-results/new-partB/L0S2/README.md
6. **Counting, field model and heatmap:** a SQLite table of the 117,687 buildings R11 found in the benchmark's 2,652 test images answers counting questions by fixed query (32.96%); the 8-bit U-Net is 33 MB and 1.43× as fast on a CPU; 73.6% of the heatmap's circled hot spots lie within 20 px of a truly damaged or destroyed building; and 1,146 of the 7B's answers are ready to train the 3B. Files: Champion_Task/H2-results/H2/README.md, Champion_Task/L1B-results/L1b/README.md, Champion_Task/T1-results/T1-v3/README.md

## Ridita Afrin Riya – Offline Report System: the Proof Run

1. **Why the agents were replaced:** in three Kaggle practice runs our 7B could not lead CrewAI's tool loop, each time in a new way: in run 2 it looped until one message reached 16,385 tokens, one over vLLM's limit, and no report was written; in run 3 it wrote a sentence *about* each report instead of the report (12 to 228 words, against Gemini's 379 to 912) and made up places; in run 4 it repeated one tool call until the step limit. Yet it answered every single question it was given.
2. **The fixed flow (10 Nov):** the fallback written down before any run: the code calls the tools in a fixed order, and the model only writes. The photo tool describes the six photos, five searches fetch the EMS-98 grade definitions, the model grades each place, the map tool labels the map, and 18 fixed guideline searches feed the 27 sections of the six reports, with no example names or grades in any prompt. Practice run 5 passed all five rules: 192 to 787 words per report, grades that follow the photos (the undamaged hall G1, three streets G4, one G5), and 11 citations in the Expert report, all real.
3. **Proof run, internet off (11 Nov):** passed all five rules. The internet was off by Kaggle's saved setting and by a failed connection test; the network log has 4 connections, all to our model's server on the same machine; no Gemini library was loaded; and the six reports are byte for byte those of run 5. On two T4 GPUs DisasTeller took 6.1 minutes (medians: 8.1 s per section, 136 ms per search). Not yet shown is that the reports are good: the map reading found "no dangerous zones" although one street is G5, one report invents a hotline number, and only the Expert report cites. Files: Offline_RAG_KR2/README.md, Offline_RAG_KR2/results/t2_proof/

## Aryan Sami – Building Query Console

1. **From BDC-v4 to a query console:** BDC-v4 showed that passing the segmenter's counts through the VLM cost 10 points, as the model ignores numbers written in its prompt, and in BDC-v3 11% of answers were sentences instead of an option. So instead of a chatbot, the web system has a fixed-question console over the building table: a responder picks a triage question ("How many buildings are in pancake collapse?", "Which structure has the highest live-void probability?", "Total destroyed buildings in Zone A") and the answer is computed from the table: no format errors, no invented numbers.
2. **Per-building attributes:** the full H2 building schema is now the front end's data model: ID, centroid, polygon footprint, floor count, footprint area, distance to the nearest destroyed building, EMS-98 grade (1–5), live trapped-person estimate and ingress vector. Selecting a building on the map shows these in a side inspector, with no model call.
3. **Offline and zero-leak audit:** queries and building inspection run entirely in the browser, with zero HTTP or WebSocket requests to outside services (no remote map server, no inference API); switching queries and filtering over 100 buildings takes under 2 ms on a standard laptop.

## Abdullah Al Noman – Project Website and Demo

1. **Design:** the website is a public showcase plus a demo where anyone can upload a before/after image pair. The wireframes (hero, benchmark matrix, dataset and method explorer, live triage workspace) were checked against the current plan: the layout and dark theme stay; the outdated login screens, ROUGE-L/RAR benchmarks and sensor-guidance tab were dropped.
2. **First version:** a React site (Vite, React Router, Tailwind) with Home, Data, Method, Results, Gallery, Demo and About pages and reusable parts (results table, before/after slider, upload box); every number and text lives in one data folder, so results are updated in one place. The demo's backend connection point (services/api.js) returns sample output until our local API is ready. Setup errors are fixed (a missing App.css import); a blank page in the browser is being debugged. Files: Noman_task/AeroRescue-site/

## Tamanna Akter Mou – New Ideas and a Bigger VLM

1. **A second look at each building:** to fix the grader's main error (67 of Turkey's 146 destroyed buildings are graded intact), a small pilot gave the grader the other buildings' scores in the image and the disaster type. Damaged-building IoU rose in all six checks, but destroyed-building IoU, our target, went both ways (−0.028 to +0.019), so it was dropped.
2. **A bigger VLM on CraftX:** tried CraftX to test a bigger VLM (Qwen3.6-27B, Qwen2.5-VL-72B in 4-bit). It does not help our project: the DisasterM3 paper shows size does not fix counting (Qwen2.5-VL 7B 34.2%, 32B 33.2%, 72B 34.8%); a 72B needs about 41 GB for its weights alone, far too big for our offline field model; and counting and grading already come from the table and our grader, not the VLM.

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## Next Week

- **Report quality:** compare the six offline reports with Gemini's July reports, then fix the map reading, made-up contact details and missing citations in a separate run.
- **Damage map:** score find-then-grade on the 2,007 unseen pairs, to check its narrow pass of the target.
- **Counting:** fixed queries over a table of the find-then-grade buildings, to close the 1.7-point gap.
- **Field model:** train the 3B on the 1,146 answers.
- **Website:** fix the blank page, add the verified numbers and gallery images, and connect the demo and the query console to our offline pipeline.